Quant Research Mock Assessment

QRT-style practice edition

Total time	82 minutes	**24 items**
Coding	1 data-science question in the companion notebook	**42 min suggested**
Theory	10 math + 5 ML fundamentals + 8 ML advanced	**40 min suggested**

Candidate _____

Date _____

Start time _____ End time _____

Independent practice material. This is an original mock, not an official or leaked Qube Research & Technologies assessment. Public candidate reports were used only to calibrate section structure and topic coverage. The invitation and on-screen rules for your actual assessment take priority.

Instructions

Read before starting

1. Start one 82-minute timer. A suggested allocation is 42 minutes for the notebook, 18 minutes for Questions 1–10, 8 minutes for Questions 11–15, and 14 minutes for Questions 16–23.
2. The companion notebook contains the single coding question. This PDF contains the other 23 questions.
3. For **single-answer** questions, choose exactly one option. For **select-all** questions, choose every correct option and no incorrect option. Suggested scoring is one point only for an exact match.
4. Unless a question says otherwise, use ordinary exact arithmetic and standard statistical conventions. A calculator is optional for this mock.
5. Do not open the `solutions/` folder until the timer ends. There is no penalty for guessing in this practice version.

Format confidence. The 82-minute duration and 1/10/5/8 counts come from the user-provided brief. A recent third-party synthesis independently reports the same four section counts, but reports 90 minutes and is not an official QRT publication. Public candidate accounts support Python, statistics, OLS, regularization, and tree-ensemble coverage while also showing role and office variation. Platform, proctoring, scoring, navigation, and calculator rules remain unverified and are not simulated here.

Public references used for calibration

- QRT official Students and New Grads page – official careers context, no assessment blueprint.
- TraderMath QRT interview guide – third-party candidate-report synthesis; same section counts, different timing.
- Glassdoor QRT interview reports and Wall Street Oasis QRT entries – anonymous, variable experiences.

Answer sheet

Q	Answer	Q	Answer	Q	Answer
1	_____	9	_____	17	_____
2	_____	10	_____	18	_____
3	_____	11	_____	19	_____
4	_____	12	_____	20	_____
5	_____	13	_____	21	_____
6	_____	14	_____	22	_____
7	_____	15	_____	23	_____
8	_____	16	_____		

Section A: Mathematics and Statistics

Questions 1–10 | 18 min suggested

Question 1 — **Single answer**

A market is in a high-volatility regime on 20% of days. A detector flags 60% of high-volatility days and 10% of all other days. Given that the detector flags today, what is the probability that today is a high-volatility day?

A. 0.20
B. 0.40
C. 0.60
D. 0.75

Question 2 — **Single answer**

Daily event counts are conditionally Poisson with unknown rate λ. The prior is $\lambda \sim \text{Gamma}(3, 2)$, using shape and rate. Over three equal observation periods the counts are 4, 2, and 5. What is the posterior mean of λ?

A. $11/3$
B. $7/3$
C. $14/5$
D. $16/5$

Question 3 — **Single answer**

Suppose $X_1, \ldots, X_n$ are i.i.d. Uniform$(0, \theta)$ and the sample mean is 4.2. What is the method-of-moments estimate of θ?

A. 2.1
B. 4.2
C. 6.3
D. 8.4

Question 4 — **Single answer**

An OLS regression with an intercept has fitted slope -0.8. The sample means are $\bar{x} = 1.5$ and $\bar{y} = 0.7$. What is the fitted intercept?

A. -1.9
B. -0.5
C. 0.1
D. 1.9

Question 5 — **Single answer**

A score is normally distributed with mean 50 and standard deviation 4. What is the z-score of an observation equal to 58?

A. 1
B. 2
C. 4
D. 8

Question 6 Single answer

What is the rank of the matrix

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 3 & 1 \end{pmatrix}?$$

A. 0
B. 1
C. 2
D. 3

Question 7 Single answer

A joint density is $f(x, y) = c(x + 2y)$ for $0 \le x \le 1$ and $0 \le y \le 1$, and zero elsewhere. What value of c normalizes the density?

A. $1/2$
B. $2/3$
C. 1
D. $3/2$

Question 8 Single answer

A chi-squared goodness-of-fit test uses seven categories. Two distribution parameters are estimated from the same data. Assuming regularity conditions, how many degrees of freedom does the reference chi-squared distribution have?

A. 2
B. 3
C. 4
D. 6

Question 9 Single answer

Let $\mathrm{Var}(X) = 4$, $\mathrm{Var}(Y) = 9$, and $\mathrm{Cov}(X, Y) = 3$. What is $\mathrm{Var}(2X - Y)$?

A. 7
B. 11
C. 13
D. 37

Question 10 Select all that apply

In the linear model $y = X\beta + \varepsilon$, which conditions are required for the usual OLS coefficient vector to be uniquely defined and conditionally unbiased given X? Assume the relevant moments exist.

A. X has full column rank.
B. $\mathbb{E}[\varepsilon \mid X] = 0$.
C. $\mathrm{Var}(\varepsilon \mid X) = \sigma^2 I$.
D. $\varepsilon \mid X$ is normally distributed.

Section B: Machine Learning Fundamentals **Questions 11–15 | 8 min suggested**

Question 11 **Single answer**

Which workflow most clearly leaks validation information into model training?

A. Split first, fit a scaler on the training set, and transform both sets with it.
B. Tune a model using cross-validation inside the training set, then evaluate once on the test set.
C. Standardize all rows first, then split the standardized data into training and validation sets.
D. Fit a feature encoder on each training fold and apply it to that fold's validation rows.

Question 12 **Select all that apply**

Which statements about linear-model regularization are generally correct?

A. Increasing the penalty strength typically increases bias.
B. Ridge regression commonly sets many coefficients exactly to zero.
C. Lasso can set some coefficients exactly to zero.
D. Numeric features should usually be put on comparable scales before applying either penalty.
E. Increasing the penalty strength typically increases estimator variance.

Question 13 **Single answer**

Only 0.5% of observations are positive, and finding positives is the primary objective. Which metric is usually more informative than raw accuracy for comparing ranking models?

A. Mean squared error
B. Area under the precision-recall curve
C. Adjusted R^2
D. Explained variance

Question 14 **Single answer**

You must validate a model that predicts next month's outcome from historical time-series features. Which split best respects the information set available at prediction time?

A. Randomly shuffle all rows before five-fold cross-validation.
B. Train on later dates and validate on earlier dates.
C. Use expanding training windows followed by later validation blocks.
D. Duplicate rare months before making a random split.

Question 15 **Single answer**

Which method is unsupervised?

A. Logistic regression
B. K-means clustering
C. Gradient-boosted classification
D. Quantile regression

Section C: Machine Learning Advanced

Questions 16–23 | 14 min suggested

Question 16

Select all that apply

Which mechanisms directly help decorrelate the trees in a standard random forest?

A. Training trees on bootstrap samples.
B. Considering a random subset of features at each split.
C. Forcing every tree to use the same best feature at its root.
D. Replacing the ensemble average with one deepest tree.

Question 17

Single answer

In gradient boosting with squared-error loss, what target does each new weak learner approximately fit?

A. The original response with no reference to the current ensemble.
B. The current residuals, equivalently the negative loss gradients.
C. The largest principal component of the feature matrix.
D. The predictions of a separately trained random forest.

Question 18

Select all that apply

Which statements about the Adam optimizer are correct?

A. It maintains exponential moving averages of first and second gradient moments.
B. It guarantees convergence to a global minimum for a non-convex neural network.
C. It creates parameter-specific adaptive step scaling.
D. It removes the need to choose a learning rate.
E. It cannot be used with mini-batches.

Question 19

Single answer

You use PCA before a classifier and estimate performance by cross-validation. Which implementation avoids leakage?

A. Fit PCA once on all data, then cross-validate the classifier.
B. Fit PCA once on all features but exclude the target column.
C. Within each fold, fit PCA on the training portion and transform the validation portion with that fitted PCA.
D. Fit PCA independently on the validation portion so it has its own components.

Question 20

Select all that apply

Two predictors are strongly correlated and contain similar signal. What can happen to ordinary single-feature permutation importance?

A. Each feature's individual importance may be understated because the other can substitute for it.
B. The importance of both features must always be overstated.
C. Permutation importance automatically removes the correlation problem.
D. Permuting the correlated features as a group is a useful sensitivity check.

Question 21 Single answer

What is the main purpose of nested cross-validation when many hyperparameters are compared?

A. Make the final model sparse by construction.
B. Use inner folds for selection and outer folds for a less biased estimate of the full tuning procedure.
C. Guarantee that the chosen hyperparameters are globally optimal.
D. Remove the need for a time-aware split in time-series data.

Question 22 Single answer

A binary classifier's scores are passed through a strictly increasing nonlinear transformation. Which statement is correct?

A. ROC-AUC is unchanged, but calibration and log loss may change.
B. ROC-AUC and log loss are both necessarily unchanged.
C. ROC-AUC necessarily decreases.
D. Calibration is necessarily unchanged because the ranking is unchanged.

Question 23 Single answer

After deployment, the marginal distribution of features remains approximately stable, but the relationship between features and the target changes. What is this most directly called?

A. Covariate shift only
B. Label shift only
C. Concept drift
D. Bootstrap instability

Stop. Record your answers, save the notebook, and end the timer before opening `solutions/answer_key.md`.